PROBLEME PROPUSE

1. Să se arate că șirul splinurilor de ordinul unu $(S_{\Delta_n})_{n\in\mathbb{N}^*}$ este un șir de aproximare care, pentru $n\geq 2$, păstrează alura funcției

Fie

$$\Delta_n = \{i/n | i = \overline{0,n}\}$$

o diviziune uniformă a $[0,1]$, $f:[0,1]\to\mathbb{R}$, continuă.

$$\begin{aligned}
S_{\Delta_n}(f)(x) &= tf\left(\frac{i+1}{n}\right)+(1-t)f\left(\frac{i}{n}\right), i=\overline{0,n-1}. \\
t &\in [0,1], x\in[i/n,(i+1)/n] : x = t\frac{i+1}{n}+(1-t)\frac{i}{n}.
\end{aligned}$$

$$\begin{aligned}
S_{\Delta_n}(e_0)(x) &= t+(1-t)=1, \\
t &\in [0,1], x\in[i/n,(i+1)/n] : x = t\frac{i+1}{n}+(1-t)\frac{i}{n}
\end{aligned}$$

$$\begin{aligned}
S_{\Delta_n}(e_1)(x) &= t\frac{i+1}{n}+(1-t)\frac{i}{n} = x \\
t &\in [0,1], x\in[i/n,(i+1)/n] : x = t\frac{i+1}{n}+(1-t)\frac{i}{n}
\end{aligned}$$

, Solution is: $nx-i$

$$\begin{aligned}
S_{\Delta_n}(e_2)(x) &= t\left(\frac{i+1}{n}\right)^2+(1-t)\left(\frac{i}{n}\right)^2 = \\
&= (nx-i)\left(\frac{i+1}{n}\right)^2+(1-nx+i)\left(\frac{i}{n}\right)^2 = \\
&= \frac{1}{n^2}\left((i+1)^2*n*x-i*(i+1)^2\right)+\frac{i^2}{n^2}-\frac{n*x}{n^2}+\frac{i^3}{n^2} \\
S_{\Delta_n}(e_2)(x)-x^2 &= -\frac{1}{n^2}\left((1-2i)nx-1+i\right)-x^2 = \\
&= \frac{1}{n^2}\left(-n^2x^2+(1+2i)nx+1-i\right) \\
\lim_{n\to\infty}\left|\frac{1}{n^2}\left(-n^2x^2+(1+2i)nx+1-i\right)\right| &= 0, \text{uniform } \hat{\imath}n \text{ raport cu } x.
\end{aligned}$$

: $-\frac{1}{n^2}((1-2i)nx-1+i)$: Fie $p_i(x)n^2 = -n^2x^2+(1+2i)nx+1-i$; $x\in[i/n,(i+1)/n]$ $\max_{x\in[i/n,(i+1)/n]}|p_i(x)| = ?$:

$$\begin{aligned}
p_i(i/n) &= 0 \\
p_i((i+1)/n) &= 0
\end{aligned}$$

$$\max_{x\in[i/n.(i+1)/n]} p_i(x) = p_i\left(\frac{2i+1}{2n}\right) = \frac{1}{4n^2}$$

2. Fie $K_n : C[a,b] \to C[a,b]$

$$K_n(f;x) = (n+1)\sum_{k=0}^{n} b_{n,k}(x)\int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t)\,dt$$

Să se arate că $(K_n)_{n\in\mathbb{N}}$ este un șir de aproximare

Ideea: P_B_K:

$$\begin{aligned}
K_n(e_0;x) &= (n+1)\sum_{k=0}^{n} b_{n,k}\int_{k/(n+1)}^{(k+1)/(n+1)} 1\mathrm{d}x = \\
&= (n+1)\sum_{k=0}^{n} b_{n,k}\frac{1}{n+1} = \sum_{k=0}^{n} b_{n,k} = 1 \\
K_n(e_1;x) &= (n+1)\sum_{k=0}^{n} b_{n,k}\left(\frac{(k+1)^2}{(n+1)^2} - \frac{k^2}{(n+1)^2}\right)\frac{1}{2} = \\
&= (n+1)\sum_{k=0}^{n} b_{n,k}\frac{2k+1}{2(n+1)^2} = \\
&= \sum_{k=0}^{n}\binom{n}{k}x^k(1-x)^{n-k}\frac{2k+1}{2(n+1)} = \\
&= \sum_{k=0}^{n}\binom{n}{k}x^k(1-x)^{n-k}\frac{k}{n+1} + \frac{1}{2(n+1)}\sum_{k=0}^{n}\binom{n}{k}x^k(1-x)^{n-k} = \\
&= \frac{n}{n+1}B_n(e_1,x) + \frac{1}{2(n+1)}
\end{aligned}$$

DEci:

$$K_n(e_1;x) = \frac{n}{n+1}B_n(e_1,x) + \frac{1}{2(n+1)} \overset{n\to\infty}{\rightrightarrows} e_1$$

Mai trebuie $K_n(e_2;x) \overset{n\to\infty}{\rightrightarrows} e_2$.

$$
\begin{aligned}
K_n\left(e_2;x\right) &= (n+1)\sum_{k=0}^{n} b_{n,k}\left(\frac{(k+1)^3}{(n+1)^3}-\frac{k^3}{(n+1)^3}\right)\frac{1}{3}= \\
&= \frac{1}{3}\sum_{k=0}^{n} b_{n,k}\left(3\frac{(k+1)^2}{(n+1)^2}+3\frac{k+1}{(n+1)^2}+\frac{1}{(n+1)^2}\right)= \\
&= \frac{1}{3}\left(\frac{n^2}{(n+1)^2}3B_n\left(e_2;x\right)+\frac{9n}{(n+1)^2}B_n\left(e_1;x\right)+\frac{7}{(n+1)^2}B_n\left(e_0;x\right)\right)
\end{aligned}
$$

Pentru arie:

$$\int_0^1 K_n\left(f;x\right)\mathrm{d}x=\int_0^1 f\left(x\right)\mathrm{d}x$$

Dar:

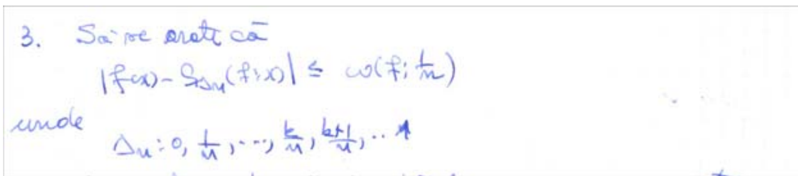
$$
\begin{aligned}
\int_0^1 b_{n,k}\left(x\right)\mathrm{d}x &= \binom{n}{k}\int_0^1 x^k\left(1-x\right)^{n-k}\mathrm{d}x= \\
&= \binom{n}{k}B\left(k+1,n-k+1\right)= \\
&= \binom{n}{k}\frac{k!\left(n-k\right)!}{(n+1)!}=\frac{1}{n+1}
\end{aligned}
$$

rezultă:

$$
\begin{aligned}
\int_0^1 K_n\left(f;x\right)\mathrm{d}x &= (n+1)\sum_{k=0}^{n}\frac{1}{n+1}\int_{k/(n+1)}^{(k+1)/(n+1)} f\left(x\right)\mathrm{d}x= \\
&= \int_0^1 f\left(x\right)\mathrm{d}x.
\end{aligned}
$$

Pentru alura: fie $F(x)$ o primitivă a lui f; SOC

$$K_n\left(f;x\right)=\sum_{k=0}^{n} b_{n,k}\left(\frac{F\left(\left(k+1\right)/\left(n+1\right)\right)-F\left(k/\left(n+1\right)\right)}{1/(n+1)}\right) >>>$$


3. Să se arate că

$|f(x)-S_{\Delta_n}(f;x)|\le\omega(f;\frac{1}{n})$

unde $\Delta_n: 0, \frac{1}{n}, \dots, \frac{k}{n}, \frac{k+1}{n}, \dots 1$

Ideea: pe $[k/n,(k+1)/n]$ funcţia f este aproximată cu coarda care trece prin $(k/n, f(k/n))$, $((k+1)/n, f((k+1)/n))$. Pentru $x\in[k/n,(k+1)/n]$ avem

$x = t\frac{k+1}{n} + (1-t)\frac{k}{n}$, $t \in [0,1]$. Atunci

$$\begin{aligned}
\left|f(x) - S_{\Delta_n}(f,x)\right| &= \left|f\left(t\frac{k+1}{n} + (1-t)\frac{k}{n}\right) - tf\left(\frac{k+1}{n}\right) - (1-t)f\left(\frac{k}{n}\right)\right| = \\
&= \left|t\left(f\left(t\frac{k+1}{n} + (1-t)\frac{k}{n}\right) - f\left(\frac{k+1}{n}\right)\right) + (1-t)\left(f\left(t\frac{k+1}{n} + (1-t)\frac{k}{n}\right) - f\right.\right. \\
&\le t\omega(f;1/n) + (1-t)\omega(f;1/n) = \omega(f;1/n).
\end{aligned}$$

6. Fie $x_{k,n} \in \left[\frac{k}{n}, \frac{k+1}{n}\right]$, $k = \overline{0,n-1}$. Să se arate că șirul $L_n(f;x) = \sum_{k=0}^{n} b_{n,k}(x) f(x_{k,n})$

Ideea:

$$|f(x_{k,n}) - f(k/n)| \le \omega(f;1.n)$$

Adică:

$$\begin{aligned}
-\omega(f;1/n) + f(k/n) &\le f(x_{k,n}) \le \omega(f;1/n) + f(k/n) \mid * b_{n,k}, \sum_{k=0}^{n} \\
-\omega(f;1/n) + B_n(f;x) &\le L_n(f;x) \le \omega(f;1/n) + B_n(f;x), n \to \infty
\end{aligned}$$

7. Să se arate că șirul $(L_n)_{n\in\mathbb{N}}$, $L_n : C[0,1] \longrightarrow \Pi_n$ definit prin

$$L_n(f;x) = \sum_{k=0}^{n} b_{n,k}(x) f\left(\sqrt{\frac{k(k-1)}{n(n-1)}}\right)$$

este un șir de aproximare care păstrează monotonia.

Vericăm că se poate reduce la pr. prec.

$$\begin{aligned}
\frac{k}{n} &\le \sqrt{\frac{k(k-1)}{n(n-1)}} \le \frac{k+1}{n} \\
\frac{k^2}{n^2} &\le \frac{k(k-1)}{n(n-1)} \le \left(\frac{k+1}{n}\right)^2
\end{aligned}$$

$$
\begin{aligned}
\frac{k^2}{n^2} &\leq \frac{k\,(k-1)}{n\,(n-1)} \\
\frac{k}{n} &\leq \frac{(k-1)}{(n-1)} \\
k\,(n-1) &\leq (k-1)\,n \\
-k &\leq -n!!!!
\end{aligned}
$$

Nu merge. Estimăm abaterea:

$$
\begin{aligned}
\left|\frac{k}{n}-\sqrt{\frac{k\,(k-1)}{n\,(n-1)}}\right| &= \left|\sqrt{\frac{k}{n}}\left(\sqrt{\frac{k}{n}}-\sqrt{\frac{(k-1)}{(n-1)}}\right)\right| = \\
&= |\sqrt{\frac{k}{n}}\frac{\frac{k}{n}-\frac{k-1}{n-1}}{\sqrt{\frac{k}{n}}+\sqrt{\frac{k-1}{n-1}}}| = \\
&= \left|\sqrt{\frac{k}{n}}\frac{\frac{kn-k-nk+n}{n(n-1)}}{\sqrt{\frac{k}{n}}+\sqrt{\frac{k-1}{n-1}}}\right| \leq \\
&\leq \left|\sqrt{\frac{k}{n}}\frac{\frac{n-k}{n(n-1)}}{2\sqrt{\frac{k}{n}}}\right| \leq \frac{1}{2\,(n-1)}
\end{aligned}
$$

rezultă:

$$
\left|f\left(\sqrt{\frac{k\,(k-1)}{n\,(n-1)}}\right)-f\,(k/n)\right| \leq \omega\left(f;\frac{1}{2\,(n-1)}\right).
$$

Se continuă raţ. de la problema 6.